

MODULE 1 : Multi-Venue Paper Corpus Collection

OpenReview

ACL
Anthology

PMLR

Preprocessing & index



63,634 papers
4 venues · 2020-2025
ACL, EMNLP, ICML, NeurIPS

Paper metadata + source PDF

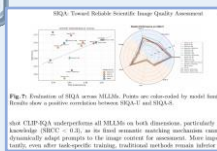
MODULE 2 : Structure-Aware Figure Extraction & Curation

Marker parser
300 DPI

1 PDF →
Structured
layout parsing

2 Caption-
anchored
figure indexing

3 Five-stage
quality curation
(S1-S5)



raw noisy
crop

clean figure

7,609 Clean
figs

MODULE 4: Evidence- Grounded 5D Scoring

Visual Clarity

Misleading
Risk

Caption
Consistency

Context
Consistency

Structure &
Layout

Score

1-10

Scientific/CS Papers



Fig3. Overall pipeline
of the method.

The results
presented in Fig.
3 further validate
the effectiveness
of our approach...

As shown in Fig. 3,
our method
significantly out
performs the base line...

MODULE 5: Benchmark Release

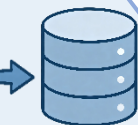
SciFigQual-



Human expert
annotation

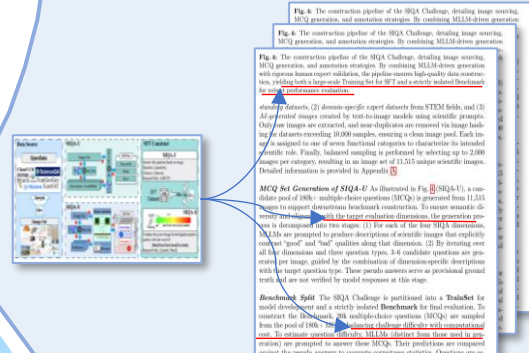


Diagnostic Panel



Benchmark
Instances

MODULE 3 : Index- Driven Context Binding



Bind text → figure, not figure alone



No Isolated figure